

“That’s a Red Flag!”: How Young Adult Instagram Users Assess Sexual Threats and Perceive Nudity Protection in Direct Messages

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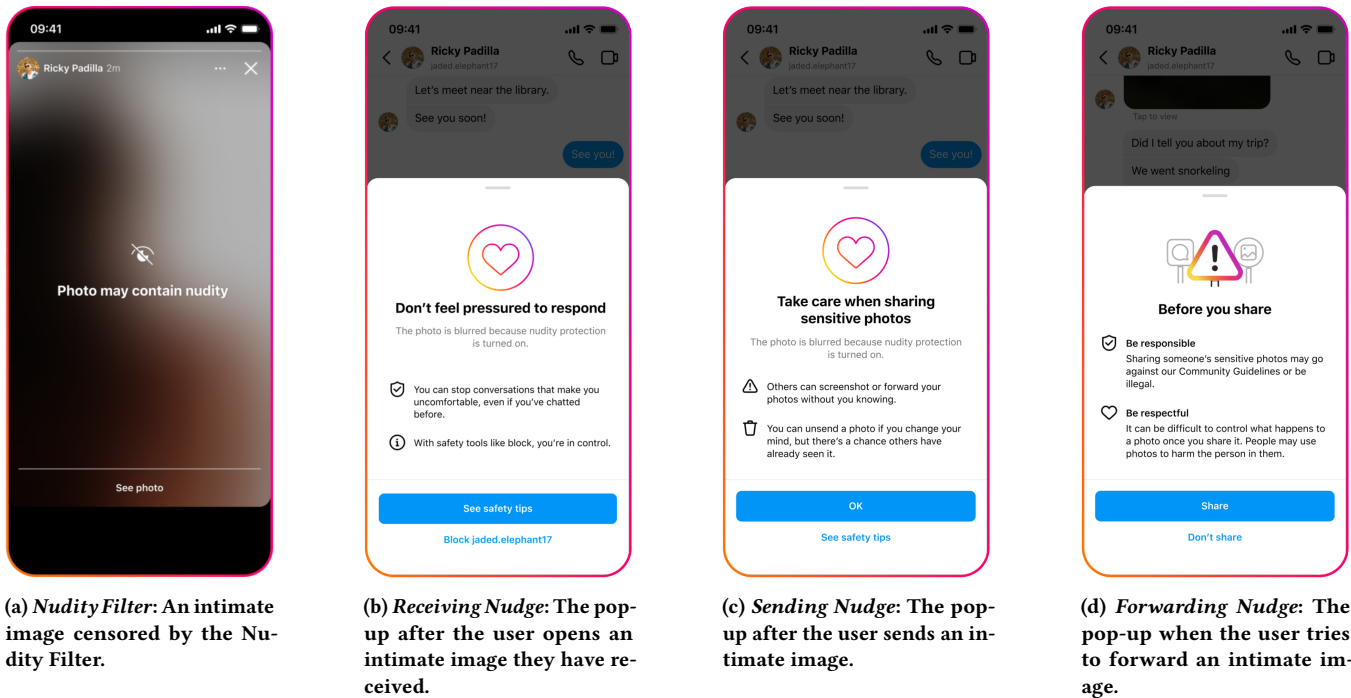


Figure 1: Instagram’s Nudity Filter and corresponding safety nudges. The images are taken from Meta’s 2024 article “New Tools to Help Protect Against Sextortion and Intimate Image Abuse” [37].

Abstract

Young adults increasingly seek intimacy and express their sexuality online. At the same time, harmful experiences such as image-based sexual abuse are also on the rise. To counter this, Meta has introduced safety tools for Instagram Direct Messages in early 2024

that blur intimate images and display informational pop-ups when receiving, sending, or forwarding them. While these safety measures exist, we do not know how they are received and if they support users’ intuitive threat evaluation. We conducted qualitative, scenario-based interviews with $n = 17$ participants aged 18 to 21 to identify detection factors for threat assessment and gather insights into users’ perceptions of these safety tools. We present the identified detection factors and suggest how this information can be used to contextualize future safety measures and guide users during risk assessment. Grounded in participant feedback, we argue for providing adequate user agency without offloading responsibility, and for greater reliability and transparency of safety tools.



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CONTENT WARNING

This paper discusses sensitive topics, including sexual threats online, which may be triggering to some.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in collaborative and social computing**; **Social networking sites**; • **Security and privacy** → **Social aspects of security and privacy**.

Keywords

Online Safety, Sexual Threats, Nudity Protection, User Threat Assessment, Instagram, Direct Messaging

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1 Introduction

Social media has become integral to many people’s lives, with Instagram being one of the most popular platforms [21, 23]. Consequently, exploring and experiencing sexuality has partially moved online, especially among emerging adults [22, 38, 48, 54]. As such, social media can provide a space to find accepting peers and experiment freely, particularly for marginalized groups, including the LGBTQ+ community [51, 54].

Yet, at the same time, harmful sexual experiences, such as image-based sexual abuse (IBSA), are prevalent online [12, 27, 34, 39, 53, 55, 60]. IBSA spans from receiving unwanted sexual solicitations to sextortion, where the perpetrator threatens to publish nude images of the victim [9, 19, 35, 48, 54, 56, 60]. The American National Center for Missing & Exploited Children reported over 26,000 sextortion cases in 2023 [39], twice the number reported the year before.

To protect Instagram users from such threats, Meta introduced safety tools – *Nudity Protection* – in early 2024 [37]. These safety tools detect and blur nude images within Direct Messages (DMs). Additionally, the tools detect opening, sending, and forwarding nude images, respectively showing different safety pop-ups. While all pop-ups provide some information, the first two additionally link to dedicated safety tips.

Similar to Meta’s tools, prior research has investigated and proposed different safety designs to protect users from harm [1, 2, 22, 32, 40, 44]. However, prior work has largely focused on gathering non-platform-specific design implications from users [22, 44] across various threat contexts (e.g., cyberbullying) [2, 32, 40]. Yet, contextualized investigations of such tools, i.e., in a specific platform and threat context, provide a more thorough understanding of users’ perceptions of these safety tools.

Furthermore, Meta’s tools are triggered when sharing nude images, thereby identifying these situations as potential indicators of harm. Similarly, prior work has explored computational methods to detect potential sexual threats in chat messages, e.g., [5, 6, 8, 47, 49]. However, considering how users themselves assess sexual threats is

crucial for gaining a complete understanding and informing safety measures. With this knowledge, safety measures can provide more targeted help in identifying threats and, consequently, offer tailored support.

To address these gaps, we investigate young adult Instagram users’ perceptions of Nudity Protection and explore their sexual threat assessment within Direct Messages. To this end, we want to answer the following questions:

RQ1: How do youth users determine whether a conversation in Instagram DMs poses a sexual threat?

RQ2: What are the perceived positive and negative design aspects of the new safety tools to help prevent online sexual threats from the perspective of youth Instagram users?

We conducted a qualitative, scenario-based interview study with 17 participants aged 18 to 21 in Germany. During the interviews, participants explored four sexual threat scenarios embedded in an Instagram-lookalike prototype. While the scenarios were fictional, the depicted threats focused on sexting and common forms of IBSA. Specifically, the scenarios covered receiving an unsolicited intimate image (cyberflashing) [28, 33], a chat potentially leading to sextortion [9, 18, 61], and non-consensual sharing of intimate images [36, 55]. Each scenario featured two or more of the discussed safety tools.

To determine the riskiness and subsequent handling of a sexual threat in DMs, we found that participants navigated a *decision flow*. Within that flow, we identified several detection factors that users evaluate for risk assessment, and we propose suggestions for how this can contextualize future safety measures. Considering Meta’s safety tools, participants argued for greater user agency while ensuring adequate user protection without offloading responsibility. We further present participants’ suggestions beyond the scope of the discussed safety tools and highlight challenges when designing for increased online safety. With this paper, we therefore make the following core contributions:

- (1) We present empirical insights on concrete detection factors for assessing sexual threats within DMs from the perspective of young adult Instagram users.
- (2) We provide empirical evidence on young adults’ perceptions of applied safety measures intended to protect from sexual threats in Instagram DMs and present resulting design implications for safety measures.

2 Background & Related Work

This section provides background on sexual threat experiences that users, including youth users such as young adults, face online. We present research on (sexual) threat detection and related work evaluating online safety measures, specifically focusing on nudge-based tools. Nudges are the basis for Meta’s *Nudity Protection*, which we present at the end of this section.

2.1 Online Sexual (Threat) Experiences

Sexting has become a normal and common part of online experiences [9, 22, 38, 48, 54]. In a study from 2022, more than 80% of 2,828 young adults reported they had sexted [52]. Sexting can be beneficial for relationship intimacy [20, 22, 55] or provide means

to explore one's sexuality [9, 51, 54]. Yet, at the same time, harmful sexual experiences are prevalent online [34, 53], also among youth [5, 9, 19, 27, 47, 48, 60]. Umbach et al. reported more than 1 in 5 participating adults from 10 countries having experienced some form of image-based sexual abuse (IBSA) [53]. IBSA encompasses several different sexual threats online, forming a continuum of gendered, sexualized forms of abuse [36]. These include receiving unsolicited intimate images (cyberflashing) [28, 33], perpetrators threatening to publish intimate images (sextortion) [46, 61], or non-consensual intimate image sharing (NCII) [35, 36]. Notably, digital technologies facilitate IBSA, for instance, by offering anonymity [7], easy access to strangers [9], or simplifying the dissemination of intimate content [31]. A large body of research suggests that perpetrators of IBSA are often known people or even intimate partners and mostly male [43, 46, 61]. However, Wei et al. found that financial sextortion was usually perpetrated by strangers and women [56]. While reasons for perpetration vary across relevant literature and IBSA subtypes, sexual gratification, and control have surfaced frequently [26, 28, 41, 56]. Previous research, therefore, underscores the need for adequate online safety measures, especially for youth users.

2.2 Detection Approaches for Sexual Threats

Prior work has focused on developing computational methods to detect sexually risky chats online, e.g., [5, 6, 8, 47, 49]. For example, Razi et al. used 150 Instagram DMs flagged by adolescents as risky to create sexual risk detection classifiers [47]. As such, they identified user age and relationship type with the chat partner as important contextual factors. Also based on risk-flagged Instagram DMs, Ali et al. investigated metadata and identified conversation length as a reliable indicator of potential threat [6]. They found risky conversations to be shorter and more one-sided than safe conversations. In contrast to computational approaches, Agha et al. identified initial factors indicating (not necessarily sexual) threats from the perspective of teens while co-designing risky scenarios [3]. While identifying threats at a technical level is crucial for automated safety interventions, the computationally identified factors may not align with those that users consider. More insights from youth users are needed to understand the factors they assess. Such insights would offer a complementary perspective to the computational classifying approaches used in related work [5, 6, 47, 49].

2.3 Evaluations of Online Safety Measures for Youth

Current work on youth online safety has shifted its focus from restrictive approaches such as parental controls towards empowering youth to build resilience against potential threats [57]. As such, nudge-based safety measures have received significant attention [1, 16]. For example, Agha et al. co-designed safety measures with teens, including informational alerts and sensitivity filters [2]. Some insights from prior work, such as the sensitivity filter, appear to be used to implement safety measures on platforms such as Instagram [37]. To understand user perception of nudge-based safety tools, Masaki et al. conducted a large-scale survey [32] and Obajemu et al. semi-structured interviews [40] with adolescents to

evaluate different concepts of nudge designs. However, these evaluations have covered a broad range of privacy and safety threats rather than focusing specifically on sexual harm [1, 32, 40]. More youth users' insights are needed, specifically focusing on safety measures protecting from sexual threats. We follow calls from Agha et al. [1] for more research evaluating safety nudges, and explore young adult users' perceptions of deployed safety measures.

2.4 Nudity Protection on Instagram

In April 2024, Meta introduced *Nudity Protection*, safety tools for young users on Instagram [37]. These tools aim to “*help protect against sextortion and intimate image abuse*” [37]. While Nudity Protection is automatically enabled only for minors, Meta states that they encourage everyone to enable it. This section provides an overview and describes these safety tools for Instagram DMs to clarify future references.

Nudity Filter. When Instagram detects nudity within a DM, the image is blurred in the chat for the sender and receiver (see Figure 1a). Additionally, an overlaid message warns that the image might contain nudity. We refer to this tool as the *Nudity Filter*. By clicking “*See photo*,” users can view the unblurred image despite the initial filter.

Receiving an Intimate Image. When users receive and open an intimate image in their DMs, a pop-up appears (see Figure 1b). We refer to this pop-up as the *Receiving Nudge*. It covers about $\frac{2}{3}$ of the image. The publicly available information [37] does not make it clear whether the users could already see the blurred image before the Receiving Nudge appears. The Receiving Nudge explains that the image is blurred since Nudity Protection is active. The pop-up reminds users not to feel pressured to respond, that they are in control, and recommends other safety tools, such as blocking the chat partner, with a single button-tap. Another button links to further safety tips we describe below.

Sending an Intimate Image. Meta has introduced a pop-up after a user sends an intimate image to someone via DMs (see Figure 1c). We refer to this pop-up as the *Sending Nudge*. The Sending Nudge explains that the image is blurred because Nudity Protection is enabled. Furthermore, it reminds the user that the image can be, for example, forwarded. It also states the possibility of unsending the image and that others might have already seen the picture. The pop-up links to further safety tips we describe below.

Forwarding an Intimate Image. If Nudity Protection is enabled, the pop-up appears when forwarding an intimate image (see Figure 1d). We refer to this as the *Forwarding Nudge*. It asks the user to be responsible and respectful when forwarding. Specifically, it reminds the user that the action might be illegal or against Meta's Community Guidelines. The Forwarding Nudge also states that what happens to the photo after sharing is beyond the user's control. The nudge allows users to share the picture or to cancel the action.

Safety Tips. Meta has implemented three pages dedicated to providing users with more information and help (see Figure 2). We will refer to these as the *Safety Tips*. The first page notes that it is hard to control images once they have been shared with others.

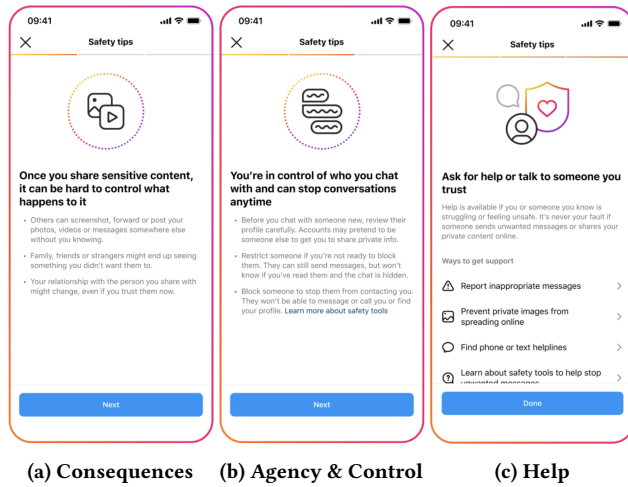


Figure 2: Safety Tips: The first of the three pages, containing further information and help. The Sending Nudge and the Receiving Nudge both link to the Safety Tips.

The second page provides tips when chatting with someone, such as checking the other person’s profile beforehand, while the last page is dedicated to getting help and reaching out to others.

3 Methodology

We conducted a qualitative scenario-based interview study with 17 young adult Instagram users to identify the factors they use to evaluate potential sexual threats within Instagram DMs (RQ1) and gain insights into how they perceive the new safety tools (see Section 2.4) introduced by Meta (RQ2). Investigating both at the same time allowed us to contextualize our results, including the acceptance and preferred timing of such tools. We did not aim for generalization, as the goal of this study was a nuanced exploration of participants’ perceptions [42]. As such, conducting interviews allowed us to thoroughly investigate participants’ perceptions of the safety tools and their sexual threat assessments.

3.1 Interview Structure

We started each interview with brief introductions and continued with initial questions to build rapport and warm up participants. During the central part of our interviews, participants explored interactive sexual threat scenarios while thinking aloud. The scenarios allowed participants to explore both the potential threat and the tools in context. We further posed questions during and after each scenario to elicit comprehensive feedback about what indicated a potential threat to them (RQ1) and extract their opinions on the presented safety tools (RQ2). We concluded each interview with overarching, more general questions about threat detection and the tools. As participants had seen the tools in various contexts, this enabled broader reflection. Afterward, participants completed a voluntary demographic information survey. Figure 3 provides an overview of the interview structure, and the full interview guide is linked to in Appendix A.

3.2 Scenarios

In total, our participants explored four different threat scenarios. Each featured at least one private chat depicting a potential sexual threat, and included two or more of the new safety tools presented by Meta (as described in Section 2.4). All scenarios showcased the use of the Nudity Filter. Additionally, scenarios A, B, and C featured the Safety Tips. Participants explored the scenarios in the following order:

Scenario A – depicted receiving an unsolicited intimate image by a stranger; a common sexual threat also referred to as cyberflashing [28, 33, 48, 60]. Here, the participants explored the Receiving Nudge.

Scenario B – showed a person building trust first, then using that trust and their power position to request an intimate image. Thus, the scenario illustrated a situation that could lead to sextortion — one of the threats Meta aims to prevent with the new safety tools [37]. This scenario included sending an intimate image, which triggered the Sending Nudge.

Scenario C – presented the context of sexting, which is common among youth [9, 22, 38, 48, 54]. This scenario again featured the Receiving Nudge. It showed the tool in a consensual image-sharing setting, allowing for direct comparison with Scenario A.

Scenario D – depicted the Forwarding Nudge in the context of an intimate image being shared without the image owner’s consent (NCII); a threat also frequently discussed in related work [36, 48, 55]. The scenario showed how an intimate image was forwarded in a group chat under peer pressure, and how the Forwarding Nudge intervened.

See Appendix A for a link to the scenario screens.

3.2.1 Scenario Creation. When creating the aforementioned scenarios, we considered threat context, related work, and the protection of participants and researchers. Choosing the scenarios’ threat context was guided by related work on encountered sexual threats [19, 35, 48, 53, 60, 61] and the threats that Meta intended to protect against [37], namely IBSA and sextortion. We also included a scenario depicting consensual sexting for participants to discuss the tools in a less threatening but common context [5, 38]. To protect both participants and researchers, we decided against using chat messages between victim-survivors and perpetrators. Instead, the first author took the threat context and created fictional DMs and chat partner profiles to provide context for participants, aiding discussions on threat assessment and the functionality of the safety tools. We intentionally varied the chat partner’s account and elements across scenarios (e.g., profile descriptions, number of posts). Finally, the scenarios were checked by two senior researchers and tested during the pilot interviews (see Section 3.3).

Ecological Validity. To increase *ecological validity*, we decided to make the scenarios interactive and look like Instagram. To achieve this, we utilized Figma.¹ Hence, participants could click through each DM, the chat partner’s profile (except for Scenario D), and the new tools. We used royalty-free images from Pexels² and Unsplash³.

¹Figma, an online prototyping tool: <https://figma.com>.

²Pexels: <https://www.pexels.com/>.

³Unsplash: <https://unsplash.com/>.

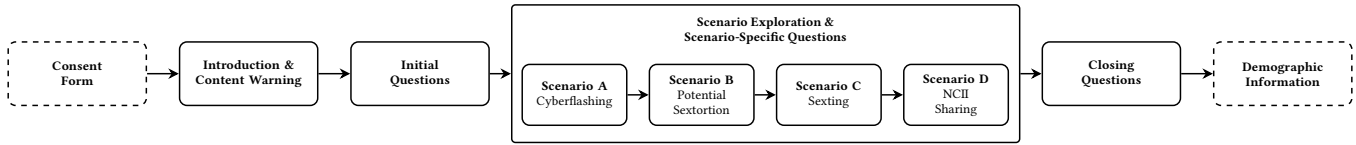


Figure 3: An overview of the interview structure. Steps with dashed lines, i.e., the Consent Form and the Demographic Information, were performed before and after the interview session with a survey tool.

Scenario Exploration. Before scenario exploration, we ensured participants were informed about the procedure while minimizing bias. We reiterated that the scenarios were entirely fictional and that the participants would not be exposed to any sexually explicit imagery. As the context of the scenarios is highly private and sensitive, ethical considerations guided every step of this work (see Appendix B). As such, we asked participants to imagine the chats were presented to them by a hypothetical friend to not have them judge their own actions. Every participant received basic information about whether and how the chat partners in the scenario knew each other (i.e., in person or just online). We included this information because prior work [19, 47] found it is important to classify the severity of risky situations, yet it is hard to extract from the scenarios on its own. Participants received no further information or guidance to prevent biasing their risk assessment. We therefore assisted participants only if they got stuck or asked for help.

3.3 Pilot Interviews

We received feedback from experienced researchers and conducted four pilot studies to refine and strengthen our interview guide (three in English and one in German). Two pilot participants were young adults, and two were doctoral researchers from our lab. Based on their feedback, we adjusted the first scenario to match Instagram’s design and shortened the introduction to avoid exceeding the interview time frame. We did not include these interviews in our analysis, as the pilot participants were not part of our target population (i.e., older than 21).

3.4 Recruitment and Participants

To best answer our research questions and conduct an in-depth exploration, we aimed to recruit a diverse set of participants. Firstly, we required participants to be between 18 and 21 years old. We recruited young adults as they can reflect on their recent teenage years and understand how previous experiences shape their threat assessment. Secondly, we only selected participants self-identifying as active Instagram users. Thirdly, related work indicates that gender and sexuality influence the kind and severity of sexual threats users are more likely to experience [51, 53, 54, 56]. We therefore screened participants based on their gender and sexual identity using Qualtrics⁴. Since Meta also translated the safety tools into German, we recruited English- and German-speaking participants to account for language and phrasing differences. We recruited participants through word of mouth and by distributing flyers in Saarbrücken and Leipzig at various locations, including university

campuses, driving schools, and youth centers. Table 1 contains an overview of participants’ demographic information.

3.5 Data Collection

We conducted interviews with 17 young adults between October 2024 and June 2025. We held the interviews online using either Zoom⁵ or Google Meet.⁶ Each interview took approximately 1 hour and 30 minutes, and participants received €20 compensation. Participants could choose whether to be audio- or video-recorded, or not at all. Ten participants consented to video recordings. Here, we captured both participants’ screens as they clicked through the scenarios and their webcams. To see what was displayed to participants and how they interacted with the prototype, we utilized the “Follow” feature in Figma. This displays the prototype screen and cursor position of the person being followed without them having to share their screen. We informed participants about this when sharing the link to the prototype. The remaining seven participants declined video recording, and one additionally declined audio recording. We therefore audio-recorded six interviews and documented one with notes. In total, we collected more than 24 hours of recorded interview data. We transcribed the recordings using a locally installed speech-to-text model and manually checked its accuracy. We also collected the participants’ demographic information and contact details using Qualtrics.⁴

3.6 Data Analysis

We applied an inductive qualitative analysis approach, following thematic analysis by Braun and Clarke [4, 14], coding iteratively and performing regular affinity diagramming. Three researchers — two doctoral researchers (junior and senior) and a postdoctoral researcher — experienced in qualitative analysis were involved. While the third researcher was not involved in the coding process, they participated in regular discussions and affinity diagramming, providing a valuable outsider perspective. As all researchers are fluent in both English and German, we did not translate any transcripts and used English codes throughout our analysis. After familiarizing ourselves with the interview data, the first author iteratively coded subsets of interviews to build a codebook. To foster a shared understanding of the data and ensure quality throughout the analysis, we regularly conducted affinity diagramming [17] of the current codebook, visually grouping the data into categories to better understand its underlying connections. To further assure quality, the second researcher independently coded one English and one German interview, after which the codes were compared, discussed, and refined. After iteratively coding and regularly discussing 12

⁴Qualtrics, an online survey tool: <https://www.qualtrics.com>

⁵Zoom, a platform for video conferences: <https://www.zoom.com>.

⁶Google Meet, a platform for video conferences: <https://meet.google.com>.

Table 1: Participant demographics. NCII = non-consensual intimate image

Interview Language	ID	Age	Gender	Sexual Identity	Country of Birth	Instagram Usage	Nudity Protection Awareness	Sexual Threat Experience	Sexual Threat
Eng	P1	21	Woman	Bi	Germany	Daily	No	Self	Cyberflashing
Eng	P2	18	Man	Hetero	Syria	Daily	No	Friends	Cyberflashing
Eng	P3	20	Man	Hetero	N/A	Daily	No	—	—
Eng	P4	21	Man	Hetero	Pakistan	Every other day	No	—	—
Eng	P5	20	Man	Hetero	N/A	N/A	No	Self	Sexual Solicitations
Eng	P6	20	Man	Bi	South Africa	Every other day	No	Friends	Sexual Solicitations
Eng	P7	21	Woman	Hetero	Germany	Daily	Yes	—	—
Eng	P8	19	Woman	Hetero	India	Daily	No	Friends	NCII Sharing
Eng	P9	20	Woman	Bi	Germany	Daily	No	Self	Sexual Solicitations
Eng	P10	18	Man	Hetero	Pakistan	N/A	No	—	—
Ger	P11	18	Woman	Hetero	Germany	Every other day	No	Friends	Cyberflashing
Ger	P12	20	Woman	Bi	Germany	Every other day	No	Friends	Cyberflashing
Ger	P13	21	Non-Binary	Bi	Germany	Daily	No	—	—
Ger	P14	18	Man	Hetero	Germany	Daily	No	Friends	Sexual Solicitations
Ger	P15	18	Woman	Hetero	Germany	Daily	No	—	—
Ger	P16	18	Woman	Hetero	Germany	Daily	No	Self	Cyberflashing
Ger	P17	18	Woman	Hetero	Germany	Daily	No	Self	Sexual Solicitations

interviews (10 in English, 2 in German), we reviewed the results and agreed on the final definitions for the top-level codes. We argue that code saturation was reached [45], as no new top-level codes emerged thereafter. The first author coded the remaining five interviews, continuously assessing whether the new data were correctly described in the codebook, followed by a final round of affinity diagramming to resolve any remaining discrepancies. We went through the data by scenario, iteratively reviewed code categories, and thereby finalized our themes. Appendix A contains a link to the final codebook.

3.7 Positionality Statement

We acknowledge that researcher demographics and experiences shape our views and, in turn, influence our work. All authors are white Europeans: the first two researchers are German, and the last two are Austrian. Three authors identify as women and one author as a man. The first author was 24 at the time of the interviews, which might have reduced perceived power distance to participants. All authors are or have been Instagram users and are familiar with its functions, which allowed us to interpret our data with more context. All researchers have studied computer science and have a background in human-centered security and privacy research, which motivated us to improve online safety without user surveillance or privacy infringements.

3.8 Limitations

While our study provides meaningful insights into young adults' threat assessment and safety tool perceptions, we acknowledge some limitations. We deliberately incorporated scenarios during our interviews to embed the tools in different threat contexts, thereby facilitating rich, contextually grounded discussions. However, they inevitably introduced some bias through their design. To mitigate this, we covered diverse threat contexts and varied the metadata included (e.g., profile elements) to enable broad discussion. We encouraged participants to consider context outside the explored

scenarios, for example, by asking if a different relationship type would change their answers and how (see Appendix A for a link to the interview guide). Overall, participants didn't seem limited by the scenarios and mentioned detection factors outside of them, e.g., tagged images. We also asked participants whether there were any scenarios where the tools might be unhelpful or harmful (i.e., not working as intended). Because our interviews were conducted in a safe testing environment, acceptance of the safety tools might have been higher than if they had been encountered during actual Instagram use. Moreover, discussing sexual experiences online is an intimate topic. While we tried to create a comfortable, judgment-free environment, participants might have been subject to social desirability bias.

4 Results

In this section, we first cover the decision flow participants navigated and the detection factors they investigated when assessing a potential threat. Understanding more about users' threat assessment helps situate users' perceptions of safety tools and helps identify possible improvements. We then present participants' perceptions of Meta's safety tools, including improvements, suggestions for future safety measures, and challenges in designing for increased online safety.

4.1 RQ1: How do youth users determine whether a conversation in Instagram DMs poses a sexual threat?

We found that participants' threat assessment and subsequent handling of a risk event followed a pattern which we refer to as the **decision flow** (see Figure 4). We provide detailed explanations for each step of the decision flow in the following subsections.

4.1.1 Risk Event. The decision flow starts with a **risk event**. A risk event is a situation that, independent of whether the user perceives it, has the potential to harm a user. Based on the scenarios

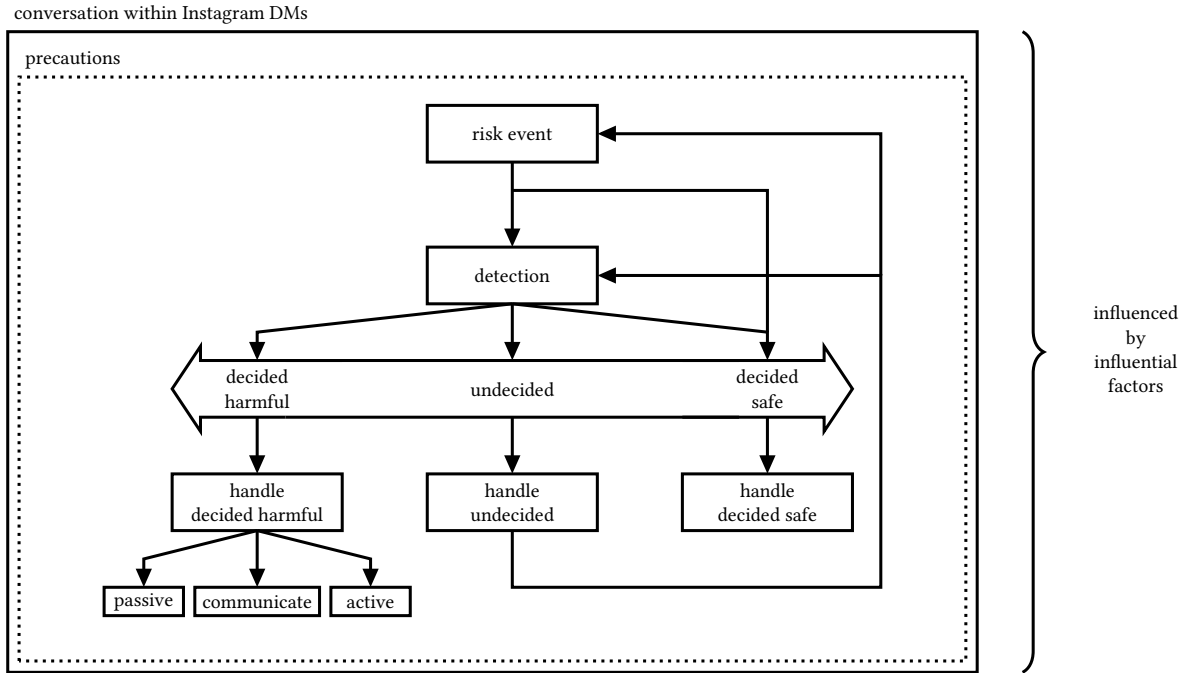


Figure 4: The decision flow for risk events within DMs.

in our interview, the risk events included receiving, sending, and forwarding intimate images. Participants mentioned that a risk event can, but does not have to, trigger a detection process. They argued that users might not identify the risk event as a potential threat, or they might not be aware that the risk event has occurred, for instance, if an intimate image of the user is being forwarded.

4.1.2 Detection. A risk event can be followed by **detection**, where users look for and evaluate detection factors. We define detection factors as indicators of harm or safety to participants. Whether a detection factor is relevant and how it affects the user’s decision-making is impacted by influential factors (see Section 4.1.5). In total, we identified four overarching categories of detection factors on Instagram: the profile, characteristics of the chat partner, messages with the chat partner, and peers’ opinions.

Profile. Participants mentioned checking the content of the chat partner’s profile when assessing the riskiness of a conversation. They were interested in what kind of images the chat partner posted: whether they uploaded “sexual picture[s]” (P2), if the images looked fake, and also how many images had been posted (“It’s most likely a spam account. They only have two images.” (P4)). Sexual content, fake-looking images, and no posts raised suspicion for a lot of our participants. It was particularly interesting for participants to check whether the chat partner had posted or shown their face in their profile picture. Usually, seeing the supposed face of the chat partner was evaluated as an indicator for safeness (“I like if the profile has

a face” (P1)), while not seeing the face of the chat partner had the opposite effect. Contrarily, P12 noted “Well, I don’t see a face, which isn’t a bad thing. I can understand that people want to protect their identity online.” P1 and P3 further took note of the visibility setting of the chat partner’s profile, i.e., whether it was set to public or private, as private profiles reveal less to non-followers. Moreover, our participants mentioned looking into whether and what stories the chat partner posted (P15), as well as “a caption that tells something about them” (P1). Participants were also interested in the interactions with others, such as comments (P6) and tagged images. Tagged images were mentioned as a positive indicator for safety, as some participants felt it increased the likelihood of an authentic profile. Participants also considered the user name of the chat partner. P15 commented, “I think the name is a little weird.” Often, participants mentioned checking the profile’s *vibe*. Participants examined whether the profile revealed “some personality” (P1). As such, they evaluated whether the profile seemed active and genuine or fake. Furthermore, participants considered both the followers of the chat partner and who the chat partner is following. Specifically, participants mentioned looking for known followers, for example, friends, or suspicious followers. P11 said, “There was someone messaging me once, and he was following tons of women, and they were all naked on their profile, in underwear and stuff.” During the scenarios, some participants also checked whether the chat partner was a follower or not. Moreover, participants evaluated the numbers of following and followers and specifically also the ratio between

them. A severe imbalance was a cause for concern for most participants. P1 said, “Well, what I first see is that he has just eight followers and he’s following many people. That’s a red flag.”

Private Messages. Participants mentioned technically identifiable detection factors in the messages exchanged between the user and the chat partner. All of these factors were noted as negatively influencing the decision regarding safety. Participants discussed the use of one-time photos (P6), receiving links, or messages with many spelling mistakes (P7). Additionally, they perceived being asked for personal information as negative. P9 mentioned, “[I] always thought people who share that much must be either really young so they don’t know about these threats, or must be older and want to have information from me.” Considering intimate image exchange, participants evaluated receiving sexual images and texts for their decision on the riskiness of a situation. Participants said that mutual intimate image exchange positively informed their decision regarding safety. P4 noted, “That’s what both sides want. We send something, and something comes back.” Similarly, approving messages from the chat partner after sending an intimate image (“She replies with ‘Wow.’” (P14)) and asking for consent were positive detection factors. Participants further considered the (perceived) intention of the chat partner, including the reason for contact or sending a specific message. For instance, if the perceived intent was to obtain an intimate image from the user (“it feels to me like he wants these pictures” (P12)), this negatively affected the users’ safety perceptions. Participants also judged the perceived behavior of the chat partner. Positive detection factors we identified included friendly, human-like messages and showing interest in the other person. P11 said, “It really seems as if Tom is just interested in getting to know my friend, so he’s flirting a little. But it’s not necessarily threatening.” Contrarily, perceived pressuring or threatening behavior negatively influenced participants’ decision-making (“that’s kind of trying to mold her mind into saying yes to [sending an intimate image]” (P8)). Participants further pointed out “weird copy-paste text” (P13) or behavior that seemed to follow a negatively perceived pattern (P14). They further rated it negatively if the chat partner commented (too much) on looks (P16) or made other inappropriate comments.

Age and Gender. When evaluating the riskiness of a situation, participants assessed available characteristics of the chat partner. Our participants considered the chat partner’s age and the age gap between chatting users. For instance, P16 mentioned, “He’s 26, so it really depends on how old my friend is. If she were the same age as me [...] I’d probably be a little cautious.” A few participants further considered the gender of the chat partner. P15 said, “I am always a bit more cautious with men anyway.”

Peer Opinions. Participants mentioned considering the opinions of their peers for their threat assessment. As such, they would ask peers for their opinion on the chat (partner). P1 mentioned, “[I] also ask friends that follow them. ‘Hey, do you know this person? Why is he contacting me?’”

4.1.3 Handling. Depending on participants’ current threat assessment, they discussed different ways to handle the situations. We grouped participants’ handling strategies into three categories: **undecided**, **decided harmful**, and **decided safe**.

Undecided. If participants were still undecided about the riskiness of a situation, they primarily mentioned gathering more information to inform their threat assessment while also taking protective measures, such as not opening or sending an image. Gathering more information included checking other social media profiles (P4) or reverse-image searching a received image to determine its authenticity (P6). Other ways included discussing the situation with peers or probing the chat partner, for example, to understand the reasoning behind their message. P17 said, “I wouldn’t block the person straight away, but would first ask them, ‘What’s that all about? Why are you doing that?’ [...] Depending on how the person responds, I would react accordingly.” This continuous evaluation indicates that threat assessment is an iterative process.

Decided Harmful. When participants decided a situation was harmful, we identified three categories of handling: **active**, **passive**, and **communicating**, categories also previously identified by Wisniewski et al. [59]. Active handling involves active measures from the user to protect themselves and, consequently, end contact with the chat partner. This includes ending the conversation, blocking, or reporting the chat partner. P14 said, “So, the only thing I can think of is to report him and just block him in any case and never write to him again.” Additionally, participants mentioned collecting evidence and deleting content. For passive handling of decidedly risky situations, a few participants discussed ignoring the potential threat (“We would just think that it is some kind of joke and to just ignore it.” (P5)). Communicative handling of a harmful situation involves contacting the chat partner’s family (P2) or discussing the risk event with the chat partner. P7 noted, “If I know the person, maybe I would still write ‘What the f*ck? Why did you do this?’” P2 and P9 further mentioned the importance of reaching out for help and support, and P2 also mentioned helping the chat partner.

Decided Safe. If participants decided a situation was safe, they mentioned continuing the conversation and, for instance, opening a received image. P5 mentioned, “I simply won’t suspect anything. This is [a] usual chat between a girlfriend and a boyfriend, and I’ll simply click on the photo because why not?”

4.1.4 Precautions. During our interviews, we found several precautions – context-independent measures – that participants applied to protect themselves online. Some of the mentioned precautions were restrictive, such as not sharing intimate images at all (“I would never share explicit images online” (P12)) or avoiding direct messages (“I try to avoid DMing people” (P2)). Participants also discussed that they “don’t text complete strangers” (P5), and mentioned using a private account. P17 said “I set my account to private so that I can see who wants to follow me and, therefore, who can send me messages.” Participants also mentioned precautions that reduce content traceability. These included strategies such as “not ever put[ting] your real face in along with anything suggestive in the same message” (P6) or sending images that disappear after opening.

4.1.5 Influential Factors. We identified several factors that can influence the whole decision flow described above. During threat assessment, they can influence if and how a user weighs and evaluates detection factors, where the thresholds between harmful, undecided, and safe lie, and, in turn, how users handle the situation. Consequently, they also influence whether a risk event triggers a

detection in the first place and what kind of precautions users take. We identified six core influential factors our participants mentioned: personal education, the user's support system, technology (such as the discussed safety tools), prior threat experiences, age, and the user's connection with the chat partner. The connection to the chat partner encompasses three subcategories. Firstly, participants considered what kind of relationship the user has with the chat partner (e.g., *"It depends on how close we are."* (P2)). Secondly, participants mentioned that trust in the chat partner is crucial (*"Well, I'd say you should definitely only do that with someone you really know and trust 100%."* (P16)). Thirdly, participants noted the importance of established mutual consent.

4.2 RQ2: What are the perceived positive and negative design aspects of the new safety tools to help prevent online sexual threats from the perspective of youth Instagram users?

We present our findings on young adult Instagram users' perceptions of Meta's new safety tools (see Section 2.4). We begin by providing a general overview of participants' perceptions, move to their suggestions for additional safety measures, and end with the challenges they identified when addressing online safety.

4.2.1 General Overview of Participant Perceptions. While participants generally appreciated the safety tools, they also identified several areas of improvement.

Acceptance. Only P7 was aware of the safety tools prior to the study. Almost all participants (except P5, P11, and partially P6) said they would enable the tools within their own Instagram accounts. Generally, they considered the new safety tools a good step towards improved online safety. P12 said, *"Now that I know they exist, I would use them, simply because I believe they are very useful and would give me a sense of security just knowing that I have these functions available."* Some participants believed the tools to be especially helpful for inexperienced users. None of our participants considered the tools harmful. At most, they were considered unnecessary, lacking new information, or annoying. Especially when sexting, P8 argued that clicking through the pop-ups would feel *"like a chore."* Another concern was that *"really nobody's reading these things"* (P5) and that users might skip through the safety tools. Participants found this particularly problematic for the Safety Tips (*"The safety tips can be helpful, but I don't really know if anyone would actually read them."* (P9)). To improve this, P1 and P9 suggested reducing text. For instance, P1 suggested that *"in this generation, where everything is with Reels and all visuals, you can maybe do a small animation or something, that shows very quickly that this content could be harmful."* Others proposed changing the page order to match the threat context (P12), forcing users to read the pages (P11), for instance, by opening Safety Tips directly (P17), and enforcing a timer before they can click away (P3). While we interviewed users both in English and German, we did not identify any significant differences in perception of the tools.

Being Protected While Having Agency. While participants argued for more safety on Instagram, they also advocated for user agency.

As such, they appreciated that the Nudity Filter protects users from seeing an intimate image immediately upon opening a chat (*"It's not like you open your phone and you see a d*ck or something"* (P7)), while at the same time allowing them to make an informed decision whether or not to view the image. P16 said, *"You are told what it is. [...] And I actually think that's pretty cool, that you have that choice, that you don't feel so helpless."* Considering the level of bluriness, some participants welcomed the ability to verify image content without fully viewing it, while P9 argued that it might still reveal too much about the image's content. Similarly, P9 appreciated that the pop-ups blocked parts of the image, yet she wished they would block the image fully. Furthermore, many participants appreciated that the Receiving Nudge offered options on what to do. They specifically liked the direct option to block the chat partner, though some participants argued that a direct reporting option would also be helpful. This wish for increased protection while preserving user agency is also reflected in participants' wish to support the user in detecting threats and implementing sharing options for (intimate) images (see Section 4.2.2). When considering younger users, however, participants tended to suggest more protection and harsher restrictions. For instance, P11 said that *"Maybe they shouldn't allow such [intimate] images at all in chats [...] belonging to people who are under 16."*

Increased Threat Awareness. The majority of participants commended the increased threat awareness provided by the safety tools from Meta. Specifically for the Receiving Nudge, participants appreciated that *"you get a warning that this might be something that you don't want to see"* (P8). P5 extended this by proposing to warn about nudity in the chat before opening the image. Considering the Sending Nudge, most participants appreciated the reminder to be careful when sharing intimate images. P13 said, *"I actually think it's quite good that this message popped up again. That way, you become aware once more of what you're sending."*

Guidance and Contextual Advice. While participants appreciated the guidance and reassurance on the tools, they argued for more contextual advice. Considering the Receiving Nudge, P15 commended that the pop-up *"gives [users] a bit of reassurance, so that they don't think, 'Yes, maybe I just find this strange, but it's normal.' Instead, Instagram reminds them that it's not normal."* Further, P12 and P14 liked that the Receiving Nudge reassured users that *"it is not your fault, that you received such content"* (P14). Many participants also noted that information on how to unsend an image and possible consequences, i.e., losing control over the sent image, on the Sending Nudge was helpful. Participants further liked that the Safety Tips offer easy access to more information, help, and reassurance. They appreciated the mention of potential consequences, for instance, *"you can no longer control what happens to it [the image]"* (P12) or *"that trust in the person may still change"* (P11). Participants acknowledged that the Safety Tips provided pointers for what to do, such as advising the user to review the chat partner's profile (*"these profile disclaimers are quite good, that you take a look at them first"* (P11)), and pointers to helplines. While participants appreciated the information above, many argued that the safety tools might feel annoying or belittling to some, and suggested providing more contextual guidance and reassurance throughout. For

instance, some participants felt that the information on the Receiving Nudge was too general and offered little new information, especially for experienced users. To improve the information on the tools, P6 suggested pointing out detection factors, e.g. *“you don’t have any mutual followers.”* Similarly, participants also recommended that stranger awareness be more prominently included in the tools, such as *“a warning like you shouldn’t send messages like this to random people”* (P2). P9 and P15 suggested highlighting that users *“do not have to engage”* (P9), and P12 emphasized that users should talk to someone about the situation. Participants also wanted the tools to point users more toward support resources. P16 wished for a concrete helpline, and P9 suggested highlighting available helplines. Furthermore, some participants proposed actively encouraging reflection through the safety tools. Suggestions included *“advis[ing] both of them to think carefully about what kind of pictures they send”* (P17) or asking *“Are you sure you want to chat with someone like that?”* (P15). A few participants suggested requiring confirmation of trust in the chat partner before sending (*“Are you sure you trust this person?”* (P9)).

Emphasizing Warning Design. Participants noted that the pop-ups looked too similar to the rest of the interface design and also worried that the nudges were too easy to ignore. P12 commented, *“Everything looks the same, which could lead me, and I think other people too, to think ‘Ah, just another security update, bye!’”* To improve this, participants recommended differentiating the warning design from the rest of the interface, for instance, by using warning icons (*“an exclamation mark”* (P5, P12)) or highlighting colors (*“red”* (P2, P3, P5, and P12)). P15 proposed making the Safety Tips button more prominent on the Receiving and Sending Nudge to garner more user attention. At the same time, P12 appreciated that the current design *“doesn’t give [the user] that feeling of, ‘Oh my goodness, [...] I need to be concerned,’ which should not be the case. It should not be too red or too colorful and overwhelming,”* highlighting that warnings need to be careful not to induce fear.

Consequence Awareness. Many participants liked that the tools highlighted consequences. Especially for the Forwarding Nudge, they appreciated that the consequences for both the potential victim and perpetrator are mentioned. Participants believed stating consequences important to deter potential perpetrators. P15 said, *“When someone reads the word illegal, they might feel a little scared because they don’t know what they could be facing, like a punishment or something.”* Still, they thought consequences needed to be addressed more directly. P9 commented, *“Writing be responsible kind of sounds like something like a teacher would say to you, and you’re like, *mockingly* ‘Oh no!’”* P7 suggested *“pointing out the risks more or how devastating the effect can be if you just share nudes from someone.”* Furthermore, participants considered informing senders about potential consequences for the receiver of intimate images. As such, P7 suggested *“check[ing] if the person [receiver] is okay with you sending a nude picture”* on the Sending Nudge.

Importance of Timing. Participants were concerned about the timing of the safety tools. P17 commented, *“It’s practically too late, for example, when you’ve already sent or received a nude photo”* for intervening. As such, the most prevalent negative aspect participants mentioned about the Sending Nudge was that the pop-up appeared

after sending. P9 thought it did not *“help that much because the person might have already seen it, so it won’t change that you’ve already sent the photo,”* and P12 mentioned the timing might even cause users to worry. To improve this, participants urged showing the pop-up before sending, and some specifically suggested adding a sending confirmation on the pop-up. Consequently, participants appreciated that the Forwarding Nudge required users to confirm forwarding within the pop-up and acknowledged that the tool was trying to prevent harm rather than help after the fact. P10 said, *“This is pretty useful actually, because it’s also giving you a wake-up call and like it doesn’t instantly share the photo.”* P12 also pointed out that confirming the forwarding in the pop-up meant *“that could make [potential perpetrators] even more accountable.”*

4.2.2 Additional Safety Tool Suggestions. Our participants provided many suggestions for safety measures that go beyond improving the discussed safety tools, ranging from prevention to detection and support.

Prevention. During the interviews, participants mentioned some safety measures before encountering a concrete risk event. They suggested implementing more stranger risk awareness, for instance, by educating users on suspicious behavior when messaging someone for the first time. More generally, participants argued for more education and support. For that, some participants proposed an always-available safety information center, and P17 advocated for educational but context-independent pop-ups. P11 suggested an educational onboarding when registering for a platform (*“You could give a brief introduction to Instagram as an app – like when people are signing up – and then point out the risks involved, such as the fact that you should be careful with sensitive content in your posts, or that strangers might message you, and so on.”* (P11)). Moreover, participants recommended improving age verification, for instance, to ensure that users under 13 could not access the platform.

Safer Image Sharing. Participants suggested different ways to make image sharing safer, suggesting greater control and flow disruption when sharing. Many participants recommended a trusted contact system, with the general idea of having fewer or no safety measures for trusted contacts. For instance, users could only share intimate images with trusted contacts. Participants further suggested measures that make it more difficult to share images without consent, thereby giving users more control over their content. As such, participants suggested setting the image-sharing default to view once to prevent unwanted saving. Other recommendations included granular per-image sharing permissions that enable users to decide, for instance, whether the chat partner can forward or screenshot an image. To increase control after sharing, participants proposed tracking image-sharing, e.g., who is sharing and who has shared originally, to better hold perpetrators accountable. P14 said, *“Instagram then kind of tracks [the image] and asks why this sensitive content [...] ended up in a different chat.”* Additionally, P6 suggested that unsharing an image should delete the image from the chat partner’s gallery, not just delete it from the chat. Some participants suggested tools that disrupt the chat flow and force a break to make safety measures less easily dismissible and provide an opportunity

to reassess a potential threat. Specifically, P3 proposed implementing a timer before being able to view an image, and P14 suggested confirming viewing an image in a separate DM with Instagram.

Detection. Participants discussed safety measures that help users detect and evaluate potential threat situations. P16 and P17 discussed flagging serial perpetrators and informing users about how many times they were reported or blocked by others. P5 suggested detecting fake profiles and highlighting both detected fake and verified profiles within the user interface. Furthermore, when messaging someone for the first time, participants proposed requiring users to confirm they know the chat partner (P12) or to review the chat partner's profile (P12), specifically their followers (P17). Participants also suggested numerous notifications to inform users about specific actions, namely forwarding, saving, and (attempted) screenshot of an intimate image. For forwarding, participants suggested notifying both the original image owner and the receiver of a forwarded image (P14).

Support. Participants offered various suggestions for increased support during and after a risk event. Participants mentioned that the system could support users when communicating with the chat partner, for instance, *"a pre-written standard message saying that you feel uncomfortable with the situation, that you would appreciate an explanation, and that you may withdraw for the time being"* (P12). More generally, P1 and P15 suggested a pop-up checking in with the user after a risk event, while P2 suggested introducing a chatbot to provide support. After a risk event, some participants argued for measures to support users in collecting evidence (*"maybe there's a way of preserving the image without like saving it to your account so that [...] the police can get to this evidence."* (P9)) or taking down images. If users reported someone, P16 suggested keeping them informed about the current progress. Considering younger users specifically, participants proposed informing parents or involving authorities.

Active Protection. Participants suggested safety measures that limit actions on platforms or impose repercussions on perpetrators to actively protect users. As such, P12 suggested contact and image requests to let users decide who can message them and who can send images. Both P12 and P6 recommended only allowing intimate image exchange when both chat partners are following each other. Some participants advised forbidding forwarding, screenshotting, and saving of intimate images. Participants further discussed measures against perpetrators, for instance, to *"ban them or just restrict"* (P9) users who are *"receiving or sending a lot of that content"* (P9). Participants voiced that social media platforms should thoroughly investigate user reports and ensure perpetrators receive *"a just punishment"* (P15). Some participants felt that enforcing rules would deter other users. P6 said that *"If you know that people have gotten banned because of stuff like this, you tread more carefully."*

4.2.3 Challenges. Participants identified several challenges when addressing online safety.

Safety Tool Limitations. Participants described several reasons users may dismiss or render the safety tools less helpful. As such, they argued that some users might be aware of a threat, but still be determined to continue the conversation. If *"the person really*

wants to send it, he's going to send it" (P2), therefore dismissing a safety measure and rendering such tools too late and ineffective. Participants explained this behavior with curiosity, naivety, or a desire for revenge. P6 commented *"teenagers aren't stupid. [...] they know it's risky, that's why they do it."* Another limitation participants pointed out was an established connection or trust with the chat partner (*"I don't need that [the Receiving Nudge], I trust this person."* (P16)). They acknowledged this was troubling since *"you could get into an argument"* (P11), resulting in a relationship change. Participants ultimately argued that it is very difficult to implement tools that are effective for everyone, and that users would likely find workarounds for safety measures, such as using a second account or saving an image and sharing it instead of directly forwarding. They also highlighted that there was a potential for misuse of safety tools, for instance, allowing perpetrators to leave less trace when sending non-consensual intimate imagery in view-once, which does not allow screenshots.

Trust in Tech. Some participants were concerned about whether social media platforms genuinely cared about user safety or whether they *"just kind of didn't want to get sued"* (P6). Furthermore, participants questioned whether Instagram's nudity detection algorithm, and thus the foundation for the safety tools, always works as intended. They were especially worried that it might not reliably detect intimate images and therefore *"malfunction and not blur some images at times"* (P4), with more participants worried about false negatives than false positives. Several participants also discussed the privacy implications of Instagram using such a detection algorithm for intimate images. P8 commented, *"Whatever picture you're sending via Instagram is getting very closely monitored, which is a bit concerning."*

Age. Participants thought that tailoring safety measures to a user's age would be beneficial. However, some participants believed that doing so is technically challenging, as with current mechanisms *"you can always fake your age"* (P1). Additionally, participants were unsure which safety measures would be best for which age bracket. Nevertheless, across participants, there was a general tendency: the younger the users, the more restrictions were suggested.

5 Discussion

Our study provides exploratory insights into users' assessments of online sexual threats and the detection factors they use to assess them. We also present a rich account of young adults' perceptions of deployed safety tools intended to protect from such threats. Based on our findings, we discuss implications for future safety measures in private messengers, propose directions for future work, and advocate for systemic changes to enhance accountability and, consequently, improve youth online safety.

5.1 Supporting Threat Detection and Evaluation (RQ1)

While the Safety Tips suggest reviewing the chat partner's profile, this information is vague and relatively hidden on page two, requiring users to click *"See Safety Tips."* In the following paragraphs, we

discuss how knowledge of user threat assessment in DMs can provide clearer information and guidance on safety measures, thereby better aiding and empowering users navigating online safety.

5.1.1 Detection Factor Awareness. Our findings show that users evaluate several detection factors when assessing the riskiness of a chat. Yet, few of these factors are currently highlighted to users, leaving them to find and interpret them on their own. This can be especially problematic when considering inexperienced users. To increase awareness of detection factors and support users in assessing threats, we propose highlighting them in the user interface, similar to how Instagram already shows the connection between chat partners at the top of each DM. For instance, a large follower-following imbalance can be visually highlighted or shown in safety pop-ups. Furthermore, detection factors within the profile are especially important when chatting with someone for the first time. Thus, we suggest advising, or even requiring, users to check and evaluate the chat partner's profile for its content, and their followers and following. Building on prior work that has explored computational threat detection [6], technically identifiable detection factors, such as links or one-time photos, could serve as triggers for future safety measures.

5.1.2 Guiding Threat Assessment. While raising awareness on detection factors is important, guiding users' threat assessments is another crucial step toward improving online safety. To assist users throughout threat assessment, we suggest safety measures that encourage users to reflect on risk events, e.g., notifications, disclaimers in the chat, or even an avatar asking users *'Did you consent to receive this image?'* Additionally, questions to guide users toward investigating and critically reflecting on detection factors can be included. For instance, *'User x is n years old. Are you comfortable chatting with someone who is n years old?'* At the same time, considering and highlighting contextual information can also affect user privacy, which we discuss in Section 5.2. Detection factors that largely positively inform safety, such as mutual intimate image exchange, also require careful consideration. For example, just pointing out mutual intimate image exchange can be problematic, as received images could be inauthentic and leveraged to trick users into sending nudes (e.g., considering patterns of financial sextortion [56, 61]). Therefore, when including guiding questions, designers need to consider malicious chat partners and not give users a false sense of safety. Consequently, guiding questions should foster reflection on the remaining threats.

5.1.3 Encouraging and Fostering Peer Support. Our participants highlighted the importance of peers not just after but also during their decision-making. We therefore argue that platforms suggest users message their (peer) support system, for instance, someone who is frequently contacted or a previously established trusted contact. Platforms can even provide a drafted message to send directly to a trusted contact. This might be especially helpful when a user is messaged by someone for the first time, to help prevent a risk event from escalating into harm. While similar suggestions were made in related work [1, 5], these focus on providing support after rather than during decision-making.

5.2 Tailoring Safety Tools (RQ2)

Participants criticized the discussed safety tools as one-size-fits-all, lacking both new information and conciseness. These criticisms echo findings from Obajemu et al. [40] and underscore the need for tailored safety measures that account for the underlying threat and unique user needs. To achieve this, we argue that including contextual information can provide more helpful and concise advice. For instance, as participants suggested, the information on the Safety Tips can be reordered to better align with the current threat context. While understanding context can improve safety tools, autonomously analyzing context can also have serious implications for user privacy. Safety measures could, for instance, analyze message content and reveal otherwise hidden information such as the chat partner's age. We therefore argue that platforms should prioritize tools that support users in detecting threats over the system autonomously detecting threats. If autonomous detection is implemented, privacy protections, such as on-device analysis, are essential. To contextualize safety tools without relying on autonomous detection, platforms could provide users with more granular controls over safety settings (e.g., when sharing intimate images) and set appropriate defaults (e.g., sharing in view-once). This gives users agency over their privacy and protection.

5.3 Balancing User Agency and Protection Responsibility (RQ2)

Our participants highlighted the importance of user agency, similar to e.g., Wisniewski et al. [58]. At the same time, participants noted that the responsibility for safety should not rest solely on the user. Our results thus show the necessity for striking a balance between user agency and platform accountability to protect users. Platform accountability encompasses both proactive measures to protect users (such as the Nudity Filter) and holding perpetrators accountable. This also aligns with related work, which suggests proactive interventions targeting the perpetrator and imposing penalties for misconduct (e.g., [1, 2, 22]). We again argue that one approach to balancing this tension is to implement appropriate default safety measures that users can adjust to some degree. Future research needs to investigate what defaults are sufficient under different contexts and how much freedom to adjust these settings youth users and safety experts deem adequate.

5.4 Reliability And Transparency of Safety Measures (RQ2)

Our findings show that young adult Instagram users are concerned about the workings and implications of the detection algorithm that triggers the safety tools. They expressed concerns about their data being analyzed and about the reliability of the analysis. A faulty detection would be especially harmful if users were actively relying on safety measures to work. However, Meta's article [37] and Instagram's settings only briefly address the detection algorithm, leaving room for questions and concerns. No information is available on exactly how the algorithm works, what data was used to train it, or whether it, for instance, detects intimate images equally well across different skin colors. Previous work on content moderation has shown how varying user groups are differently affected, including how overzealous content moderation disproportionately

affects transgender users or women [10, 24]. We briefly tested Nudity Protection by creating dummy accounts and sharing intimate images from sexual education classes showing nude female and male anatomy. Similar to reports online (e.g., [13, 25]), we found the tools unreliable, for instance, not displaying the nudges when forwarding or sending. Prior work has also highlighted the importance of communication about safety tools [30] and that transparency can increase trust [29]. We therefore argue that platforms first need to ensure that safety measures function reliably and, second, be more transparent about how they analyze and handle user data to support those measures. To improve transparency, platforms could, for instance, let independent instances test the effectiveness of safety measures or provide a safe place for users to do so.

5.5 Privacy, Protection and Profit (RQ1, RQ2)

When designing for increased online safety, goals such as privacy and safety can be in tension and thus require balancing. Platforms could, for instance, introduce tailored safety measures by analyzing user data or context. While this may increase user safety, it can also affect user privacy. Additionally, imposing harsh restrictions could lead users to move to other platforms, which is not in the interest of companies that profit from keeping users on their platforms. Given the resulting conflict, we argue that platforms, even with users' best interests at heart, cannot objectively address user safety. This calls for structural changes. Prior work has already advocated that policymakers enforce change and foster collaboration among stakeholders [15]. While we agree that such policies are necessary, we urge the establishment of an independent oversight body. It should review the implementation of such policies and decide on concrete measures. For example, user reports could be sent to this independent instance for review and assessment of further steps, such as whether to ban a user for misconduct. As this would require access to sensitive data, careful consideration of user privacy is paramount; for instance, access to data should be kept to a minimum. Moreover, this independent body could assess the reliability of safety measures, such as Nudity Protection. Consequently, having such an instance might increase user safety, trust, and fairness on social media platforms. Future work could explore potential privacy mitigations and user acceptance of such an approach.

6 Conclusion

Despite the attention youth online safety has received, we still lack safety measures that effectively protect from sexual threats without resorting to harsh restrictions that limit user autonomy and expression. While Nudity Protection for Instagram Direct Messages is an important step towards improved online safety, young adult users question its effectiveness and reliability. Efforts are needed to support users in addressing, but also detecting, threats. Above all, we argue for change that involves and empowers all users – ensuring they are heard, supported, and protected. For this, users need to be given adequate agency over safety measures, without offloading all responsibility on users. Additionally, safety measures must be just and communicated transparently.

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A Open Science

Following open science efforts, we provide our artifacts to improve the reproducibility and replicability of our research. You can find all relevant documents listed below under <https://doi.org/10.5281/zenodo.22546670>:

- Interview Guide
- Final Codebook
- Screening Survey
- Consent Form
- Demographic Information Survey
- Demographic Information Overview (see Table 1)
- Recruiting Flyer
- Interview Scenarios

We do not provide the interview transcripts to protect the rights of our participants, especially since we covered personal and sensitive topics during the interviews. Participants consented only to a few researchers analyzing their data, not publishing it in its raw form. Fully anonymizing the transcripts would additionally render the data considerably less meaningful.

B Ethical Considerations

Due to the sensitivity of our research topic, ethical considerations guided every step of the research process. Firstly, our university's ethical review board (ERB) approved our study. We fully informed all participants about the study's purpose, procedure, and their right to withdraw at any time, and participants provided written consent. We ensured that all data was stored in accordance with GDPR guidelines, and only the authors of this paper had access to unanonymized data.

The goal of this work is to empower youth, which is why we included them in our research and proposed safety measures that align with their views and needs. We recruited participants who were at least 18 years old to ensure they could consent legally. We decided to interview young adults first to gain their insights and develop a better understanding of the problem space before conducting research with minors. To protect our participants, we followed recommendations from prior work [3, 11, 50]. For instance, during the recruitment, in our consent form and at the beginning of each interview, we included content warnings for sexual content. Additionally, we included help lines and support pages on the consent form for our participants. We aimed to establish a judgment-free zone during each interview, allowing participants to be comfortable and get their valuable, honest opinions. For example, introducing the scenarios from a friend's perspective was a deliberate decision to prevent the participants from having to judge their own actions. Moreover, we frequently checked with participants to ensure they were comfortable continuing and included the opportunity for breaks or ending the interview at any time without needing to provide a reason. Further, we did not include any intimate images in the presented scenarios; instead, we used only blurred images of human-like sculptures. Therefore, we did not require participants to bypass the blur of the Nudity Filter by clicking on “*See photo*.” If they decided to click “*See photo*,” they could only see a description of the image content with the image itself still blurred. We implemented this to protect them from seeing intimate images while still conveying context to showcase realistic scenarios.

Considering the researchers, we performed regular check-ins and discussed our mental health, also reiterating the importance of breaks from the project, if needed. We also ensured that every researcher was comfortable with their tasks. For instance, we only briefly tested the tools for Nudity Protection with images we felt comfortable with (i.e., intimate images from a human biology class) and messaging testing accounts.

Further, we believe the results of this work will benefit not only young adult users on social media platforms but also users of all ages. While publishing this paper might affect Meta's reputation and finances, we argue that the benefits of improving youth online safety outweigh this.